Hasan Taha Bagci

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Education

Istanbul Technical University Sep. 2021 - Jan. 2027
B.Sc. in Artificial Intelligence and Data Engineering *Istanbul, Turkey*
Research interests: Physical AI, robot learning, safe AI, visual navigation, and model-based planning.

Publications

[1] **H. T. Bagci**, J. Bi, F. E. Yalcindag, X. Zhou, E. Koyuncu, and Z. Cao. *Safe Planning for Contact-Rich Robotic Manipulation*. Under review at the International Conference on Learning Representations (ICLR 2027).

[2] **H. T. Bagci**, E. Koyuncu, and N. K. Ure. "CONCORD: Constraint-aware decision support framework for tactical air COA generation and ranking". In: *AIAA SCITECH 2027 Forum*. Accepted; to appear. 2027.

[3] A. Şanlan, E. C. Altunkaya, **H. T. Bagci**, E. Koyuncu, and İ. Özkol. *TANGO-VIO: Triangulation-Aware Navigation with Guaranteed Feature-Observability for Visual-Inertial Odometry*. Under review at IEEE Transactions on Robotics (T-RO). 2026. arXiv: 2608.02079 [cs.RO].

[4] A. Şanlan, **H. T. Bagci**, and E. Koyuncu. "Particle Filter-Based Localization Using Visual Feature Synchronization in GNSS-Denied Navigation". In: *2025 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. 2025, pp. 1–8. DOI: 10.1109/ICNS65417.2025.10976891.

Research Experience

Constrained World-Model Planning for Robotic Manipulation Jun. 2026 - Present
Independent Research Project

- Collaborated with researchers at MIT and Nanyang Technological University on world-model planning for contact-rich robot assembly.
- Developed a closed-form action projection method that uses the live controller state to enforce force limits without online optimization.
- Evaluated the method on peg insertion, gear meshing, and nut threading in Isaac Sim and Isaac Lab FORGE; achieved 98.1% safe success on peg insertion under a 5 N force limit.

Boston University Robotics Lab Aug. 2025 - Present
Visiting Researcher; Advisor: Prof. John Baillieul *Boston, MA, USA*

- Developed a monocular depth-estimation pipeline from time-to-transit cues, using ORB features and Lucas-Kanade optical flow with yaw compensation.
- Exploring diffusion-based navigation from estimated depth and expert demonstrations, with closed-form action constraints for safe navigation.

ITU Aerospace Research Center Sep. 2024 - Present
Researcher; Advisor: Assoc. Prof. Emre Koyuncu *Istanbul, Turkey*

- Developed vision-based small-object detection and tracking methods for locating UAVs in onboard camera footage as part of the Drone Interceptor project.
- Built and tested simulation-to-flight workflows with ROS and Gazebo, including geofencing and safety checks.

MIT McGovern Institute May 2025 - Oct. 2025
Research Intern, Senseable Intelligence Group *Remote*

- Contributed to Riverst (GitHub), an open-source platform for real-time avatar conversations, working on speech-driven interactions, gestures, and lip synchronization.

Honors and Awards

2025 **Best of Session Award**, ICNS Conference.

2024 **2nd Place (Team)**, RoboNation SUAS Competition.

2023 **1st Place (Team)**, RoboNation SUAS Competition.

2021 **Ranked 2,831st nationwide (top 0.1%)**, Turkish University Entrance Examination (YKS), among approximately 2.5 million candidates.

Industry Experience

BAYKAR Defence Technologies Oct. 2024 - Dec. 2024
Intern, Artificial Intelligence and Software Development Department *Istanbul, Turkey*

- Worked on an AI-based tactical command-and-control framework, contributing to the development of decision-support software.

TUBITAK Aug. 2022 - Sep. 2022
Intern, Scientific and Technological Research Council of Türkiye *Kocaeli, Turkey*

- Gained experience in multi-object detection and tracking by implementing and testing computer vision methods.

Academic Affiliations and Service

MIT Sponsored Guest, Massachusetts Institute of Technology: htb@mit.edu.

Visiting Researcher, Boston University: htbagci@bu.edu.

Reviewer, IEEE International Conference on Robotics and Automation (**ICRA 2027**).

Member of IAIFI, NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

Leadership, Teaching and Volunteering

ITUNOM UAV Team, Istanbul Technical University Sep. 2022 - Sep. 2024
Team Leader and Software Subteam Member *Istanbul, Turkey*

- Led a 24-member UAV team, coordinating work across software, avionics, and mechanical subteams.
- Contributed to the team's perception, localization, and control software.

Educator and Mentor, T3 Foundation (Dec. 2022 - Nov. 2024): led hands-on STEM activities for middle and high school students.

Python and AI Instructor (Oct. 2022 - Jan. 2023): taught introductory programming and machine learning to high school students.

Earthquake Relief Volunteer, Turkey (Feb. 2023): helped provide educational support and essential supplies for children in the Kahramanmaraş earthquake-affected region.